Describe a wave

antocciphysics | TEACHER KEY | CPA

Learning target: Identify amplitude, wavelength, frequency and period.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 11; 3 course days later.*
15°C; ΔT is proportional to absorbed energy under these conditions.

02. Explain a worked example (worked)
Amplitude vertical from equilibrium, wavelength horizontal crest spacing; period belongs on time axis.

03. Look and explain (model)
Crest-to-trough is 2 A for a symmetric sinusoid; wavelength matches successive crests.

04. Build the solution (practice)
2 Hz;.5 s.

05. Use the idea (reason)
No; maximum displacement and cycles per second are different independently variable properties in the basic model.

06. Choose a method (mixed) *Review: Unit 5, Day 2; 12 course days later.*
$Q=mc\Delta T=19200$ J, assuming the supplied energy heats the sample.

07. Exit check (exit)
3 cycles;.5 Hz.

08. STRETCH: Apply it another way (stretch)
Vertical excursion differs, crest spacing unchanged. Amplitude alone does not determine frequency.

Describe a wave

antocciphysics | TEACHER KEY | HONORS

Learning target: Identify amplitude, wavelength, frequency and period.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 11; 3 course days later.*
Absorbed 7500 J; source 150000 J.

02. Explain a worked example (worked)
Amplitude vertical from equilibrium, wavelength horizontal crest spacing; period belongs on time axis.

03. Represent and interpret (model)
Crest-to-trough is 2 A for a symmetric sinusoid; wavelength matches successive crests.

04. Solve and justify (practice)
25 Hz;.040 s;.040 m amplitude.

05. Analyze the reasoning (reason)
No; maximum displacement and cycles per second are different independently variable properties in the basic model.

06. Choose a method (mixed) *Review: Unit 5, Day 2; 12 course days later.*
$c=Q/(m\Delta T)=1000$ J/(kg·°C). No phase change and no unaccounted loss/input.

07. Exit check (exit)
$\lambda=.50$ m; f 10 Hz.

08. STRETCH: Apply it another way (stretch)
f=2.5 Hz; T=0.4 s; v=3 m/s.